TSA Forecast Competition Final Report

https://github.com/maevegr/GualtieriReed_Kasten_Zhang_ENV797_TSA_ForecastCompetition_S26

Maeve Gualtieri-Reed, Sadie Kasten, Leo Zhang

2026-04-22

Introduction

This report forecasts July 2011 electricity load using daily electricity load, temperature, and humidity data from January 1, 2005 to June 30, 2011. It uses a training window of January 1, 2005 to June 30, 2010 to forecast and test models compared to the test window of June 30, 2010.

Full code and testing done by each report contributor can be found in the GitHub repository: https://github.com/maevegr/GualtieriReed_Kasten_Zhang_ENV797_TSA_ForecastCompetition_S26

ChatGPT was used to learn how to add exogenous variables to ARIMA + Fourier and Neural Network. It was also used for general debugging.

Methods

The load data was uploaded and converted to daily data. It was converted to a time series using msts().

```
load <- read_excel("load.xlsx") %>%
  mutate(daily = rowMeans(across(3:26), na.rm = TRUE)) %>%
  mutate(date = ymd(date)) %>%
  select(date,daily)
```

```
msts_load <- msts(load$daily,
                 seasonal.periods =c(7,365.25), #7 = weekly, 365.25 = yearly
                 start=c(2005,1,1))
```

A training and testing section of the data were separated out in order to estimate model suitability before uploading into Kaggle

```
ts_load_train <- subset(msts_load,
                        end = length(msts_load)-366)

ts_load_test <- subset(msts_load, start = length(msts_load)- 365,
                        end = length(msts_load)-335)
```

The data was examined to determine what types of models might be most suitable for the data. It is clear there are strong multiple seasonalities which indicates ARIMA + Fourier and TBATS will be good candidates. ETS + STL may not be a great model because it seems that the seasonality is not perfectly stable as indicated by the persistence of some kind of pattern in the decomposition residuals. Neural network will also be explored to see if there are non-linear patterns that are better predicted through a learning model.

```
autoplot(msts_load)
```

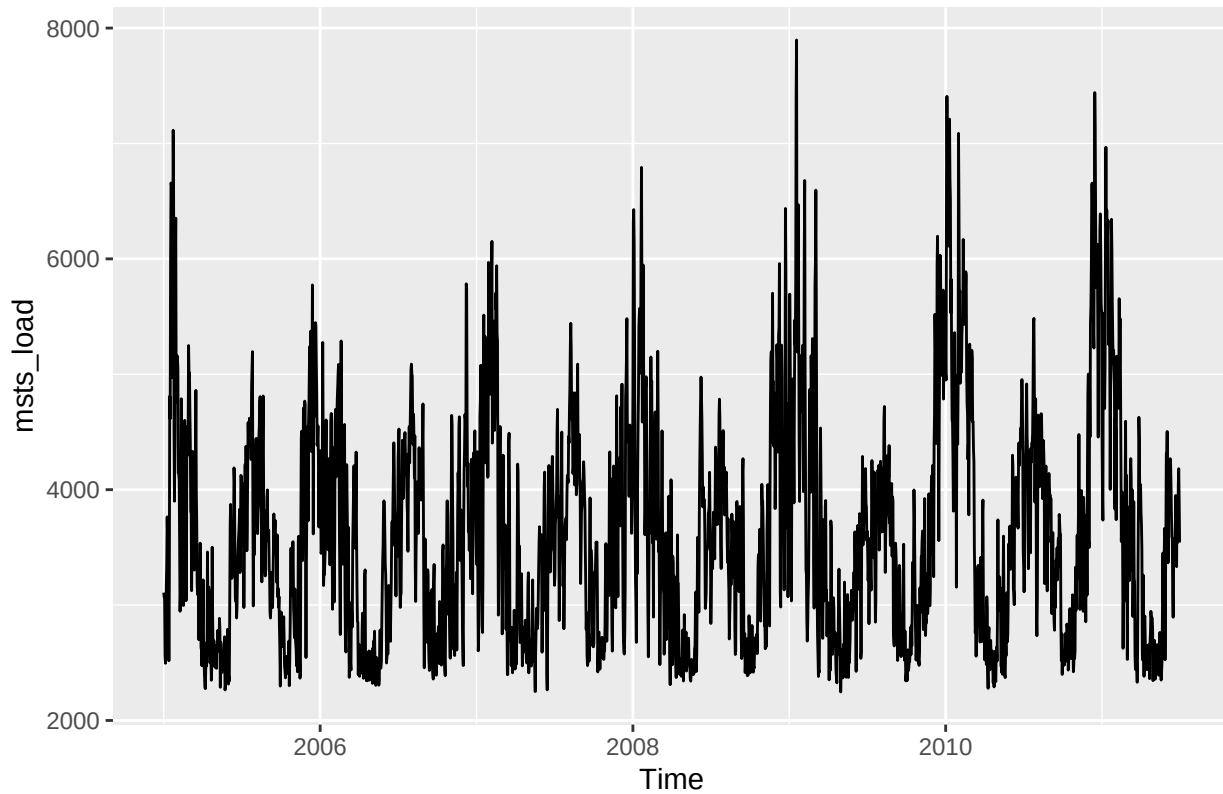

Figure 1: Plot of Raw Data

```
decomp <- msts_load %>% mstl()
autoplot(decomp)
```

Exogenous variables were loaded and turned into vectors. To be used successfully in the model forecast, forecasts of the exogenous variables also had to be constructed. This was done with the TBATS model after comparison between ARIMA and TBATS on one forecasting model proved better results via the TBATS forecast of the variables.

```
temp_df <- read_excel("temperature.xlsx") %>%
  mutate(hourly_temp = rowMeans(across(3:26), na.rm = TRUE)) %>%
  mutate(date = ymd(date)) %>%
  group_by(date) %>%
  summarise(daily = mean(as.numeric(hourly_temp), na.rm = TRUE)) %>%
  select(daily) %>%
  na.omit(temp)

temp <- temp_df$daily #vector

temp_train <- temp[1:(length(temp) - 366)]
temp_test <- temp[(length(temp) - 365):(length(temp) - 335)]
```

```
humidity_df <- read_excel("relative_humidity.xlsx") %>%
  mutate(hourly_humidity = rowMeans(across(3:26), na.rm = TRUE)) %>%
  mutate(date = ymd(date)) %>%
  group_by(date) %>%
  summarise(daily = mean(as.numeric(hourly_humidity), na.rm = TRUE)) %>%
```

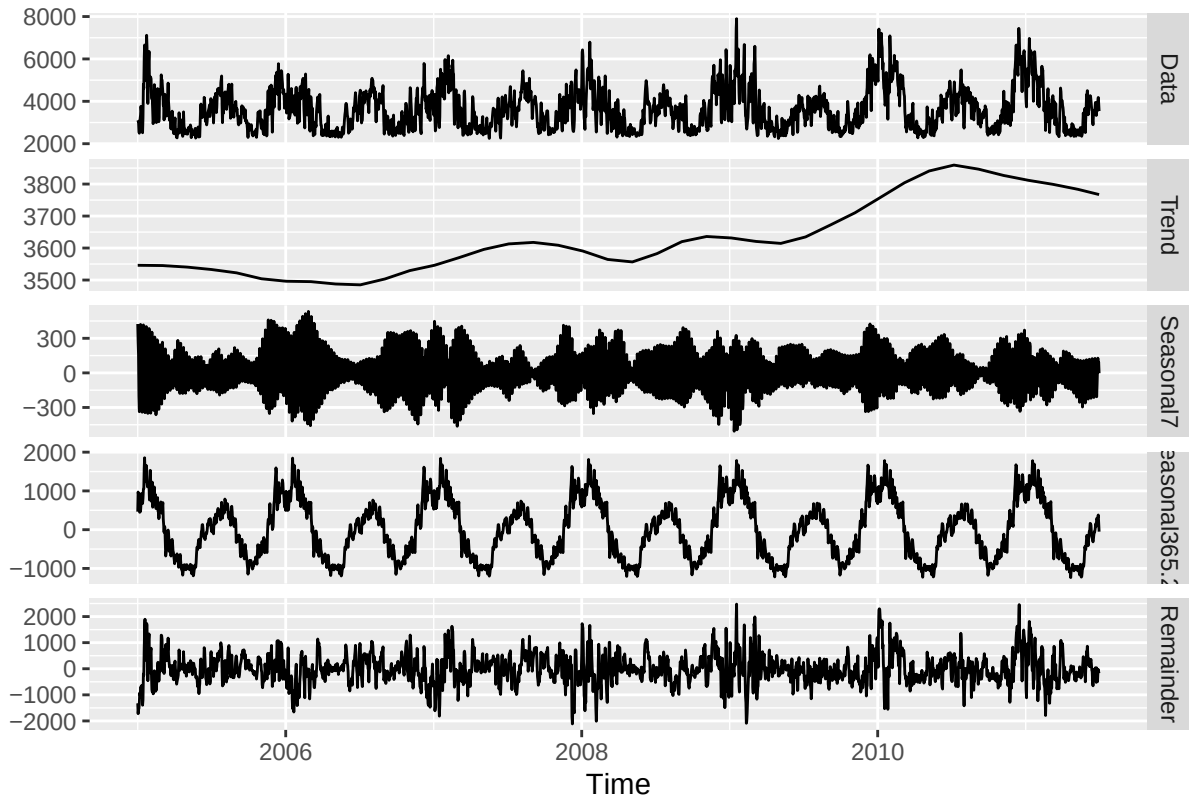

Figure 2: Decomposition

```
  select(daily) %>%
  na.omit(humidity)

humidity <- humidity_df$daily #vector

humidity_train <- humidity[1:(length(humidity) - 366)]
humidity_test <- humidity[(length(humidity) - 365):(length(humidity) - 335)]
```

```
temp_July <- forecast(tbats(temp), h = 31) #forecasting future 31 from the vector
humidity_July <- forecast(tbats(humidity), h = 31) #forecasting future 31 from the vector
```

Exogenous variables were tested via correlation tests to determine if both should be included in the forecasting models. Based on correlation tests and testing models with and without each variable, both exogenous variables were used, if used in a forecasting model.

```
cor(msts_load, temp)
```

```
## [1] -0.470026
```

```
cor(msts_load, humidity)
```

```
## [1] -0.1952849
```

```
cor(temp,humidity)
```

```
## [1] 0.3701806
```

Model selection was determined by which were forecasting models would be best suited to handle load data

and exogenous variables if used. The TBATS, ARIMA + fourier terms, ETS, and Neural Network forecasting models were all selected as testing candidates.

Results

The top 5 models are:

1) ARIMA + Fourier (K = 2,12) with temperature and humidity as exogenous variables
2) ARIMA + Fourier (K = 2,10) with temperature and humidity as exogenous variables
3) TBATS
4) Neural Network (K = 2,1) with temperature and humidity as exogenous variables
5) Neural Network (K = 2,12) with temperature and humidity as exogenous variables

Additional models, including ETS + STL, ARIMA + Fourier and NN without exogenous variables, as well as multiple attempts with different K values were tested. ETS+STL resulted in a MAPE score of 0.1330, which is better than the seasonal naive, but not as good as the ARIMA, TBATS, or NN.

TBATS

The first of the five top models that performed best against the actual data for July 2011 was created using the TBATS forecasting model. This model was trained using the simply the load data and forecast for the month of July. The MAPE score of this forecasting model was 0.0998. The forecast result is plotted below.

```
#TBATS Training Forecast & Test Window

TBATS_train <- tbats(ts_load_train)

TBATS_train_forecast <- forecast(TBATS_train, h = 31)

accuracy(TBATS_train_forecast$mean, ts_load_test)
```

```
##               ME     RMSE      MAE       MPE     MAPE      ACF1 Theil's U
## Test set -92.435 554.7923 444.1679 -4.088589 11.21313 0.6355816  1.244326
```

```
#TBATS July 2011 forecast
TBATS_model <- tbats(msts_load)

TBATS_forecast <- forecast(TBATS_model, h=31)

plot(TBATS_forecast)
```

```
TBATS_data<- data.frame(
  date = seq(as.Date("2011-07-01"), length.out = 31, by = "day"),
  value = TBATS_forecast$mean
)

#write_csv(TBATS_data, 'Submissions/TBATS_data.csv')
```

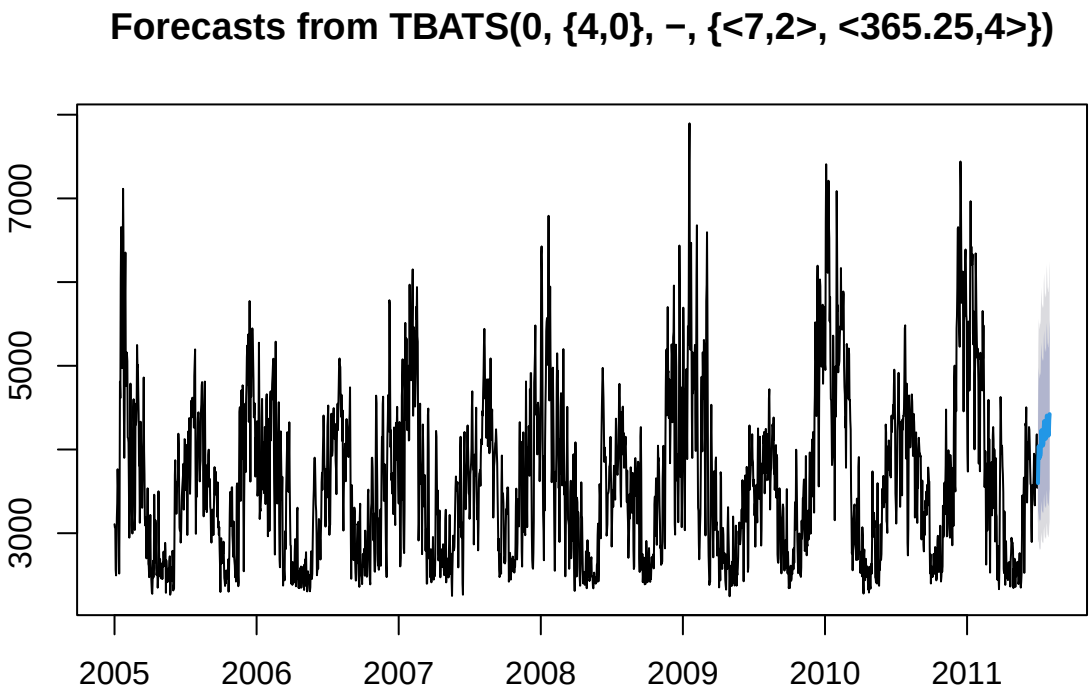

Figure 3: TBATS Forecast

Neural Network

Below are the two Neural Network models that also ranked among our top 5 forecasts based on actual data.

The first Neural Network model includes exogenous variables for temperature and humidity as well as exogenous K values 2 and 12 after testing multiple versions of K value options. The MAPE score for this forecast was 0.1169 after optimization.

```
# NN K = 2,12 Training Forecast & Test Window

#start with train window
temp<-temp_train
humidity<-humidity_train

#training model
NN_train1 <- nnetar(ts_load_train,
                p=10,
                P=0,
                xreg= cbind(fourier(ts_load_train, K=c(2,12)), temp, humidity))

#change to test window
temp <- temp_test
humidity <- humidity_test

#forecast with train model
NN_train_forecast1 <- forecast(NN_train1,
                   xreg= cbind(fourier(ts_load_train, K = c(2,12), h = 31),
                               temp,
                               humidity))

accuracy(NN_train_forecast1$mean, ts_load_test)
```

```
##                ME     RMSE      MAE        MPE     MAPE      ACF1 Theil's U
## Test set 34.4973 327.5359 251.6464 0.06481556 6.043572 0.3061996 0.6835599
```

```
## NN K = 2,12 July 2011 forecast
NN_4 <- nnetar(msts_load,
                p=10,
                P=0,
                xreg= cbind(fourier(msts_load, K=c(2,12)), temp, humidity))

NN_4_forecast <- forecast(NN_4,
                  xreg= cbind(fourier(msts_load, K = c(2,12), h = 31),
                              as.numeric(temp_July$mean),
                              as.numeric(humidity_July$mean)))

plot(NN_4_forecast)
```

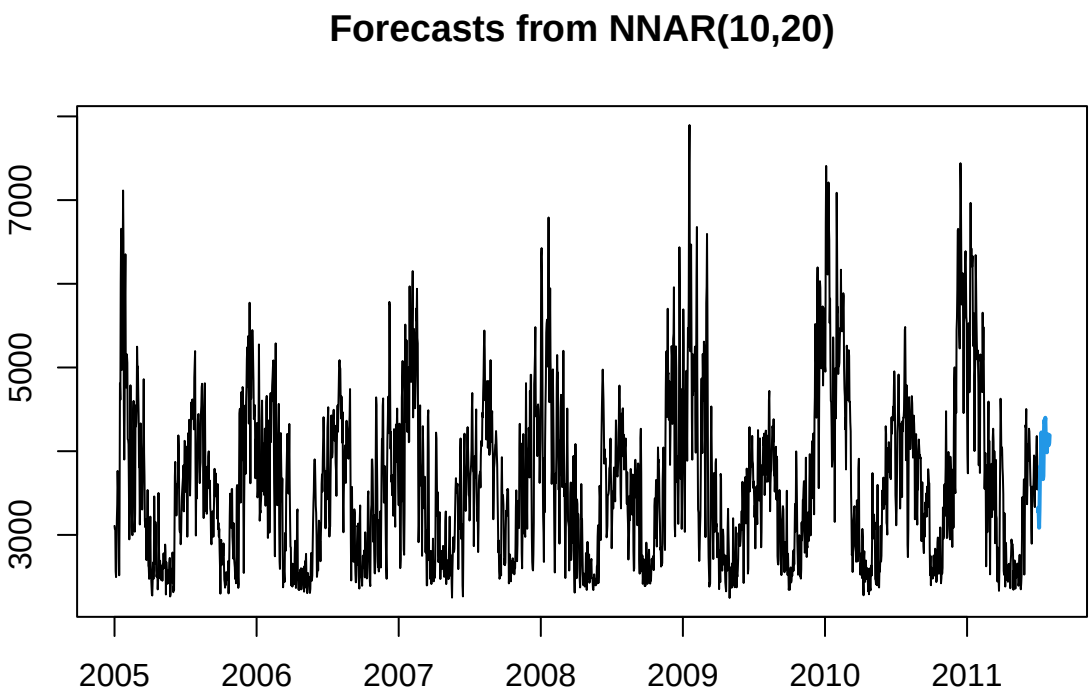

Figure 4: Neural Network, Version 1

```
NN_4_data<- data.frame(
  date = seq(as.Date("2011-07-01"), length.out = 31, by = "day"),
  value = NN_4_forecast$mean
)

#write_csv(NN_4_data, 'Submissions/NN_4_data.csv')
```

The second Neural Network model also includes exogenous variables for temperature and humidity as well as exogenous K values 2 and 1 after testing multiple versions of K value options. The MAPE score for this forecast was 0.1134 after optimization.

```
## NN K = 2,12 Training Forecast & Test Window

#start with train window
```

```r
temp<-temp_train
humidity<-humidity_train

#training model
NN_train2 <- nnetar(ts_load_train,
                p=10,
                P=1,
                xreg= cbind(fourier(ts_load_train, K=c(2,1)), temp, humidity))

#change to test window
temp <- temp_test
humidity <- humidity_test

#forecast with train model
NN_train_forecast2 <- forecast(NN_train2,
                  xreg= cbind(fourier(ts_load_train, K = c(2,1), h = 31),
                              temp,
                              humidity))

accuracy(NN_train_forecast2$mean, ts_load_test)
```

```
##                ME     RMSE      MAE        MPE     MAPE      ACF1 Theil's U
## Test set 28.03814 328.9699 249.6003 -0.1964913 6.027405 0.3781043 0.6768152
```

```r
NN_6 <- nnetar(msts_load,
                p=10,
                P=1,
                xreg= cbind(fourier(msts_load, K=c(2,1)), temp, humidity))

NN_6_forecast <- forecast(NN_6,
                  xreg= cbind(fourier(msts_load, K = c(2,1), h = 31),
                              as.numeric(temp_July$mean),
                              as.numeric(humidity_July$mean)))
plot(NN_6_forecast)
```

```r
NN_6_data<- data.frame(
  date = seq(as.Date("2011-07-01"), length.out = 31, by = "day"),
  value = NN_6_forecast$mean
)

#write_csv(NN_6_data, 'Submissions/NN_6_data.csv')
```

ARIMA + Fourier Terms

Similar to the Neural Network models, the ARIMA models that ranked best compared to the actual data also included exogenous variables for temperature and humidity.

Our highest performing ARIMA model, and model overall, included fourier terms where K= 2 and 12 and temp and humidity forecasts were included. The MAPE score for this forecast was 0.0965 after optimization.

Initially, K = 2,12 was used as a baseline estimate for the best K values. Looking at the test window of July 2010, this model does not perform very well compared to the other models. However, using the full dataset to train and forecast July 2011, this model is our best performing with the lowest MAPE.

```r
## ARIMA +FOURIER (K = 2,12) Training Forecast & Test Window
```

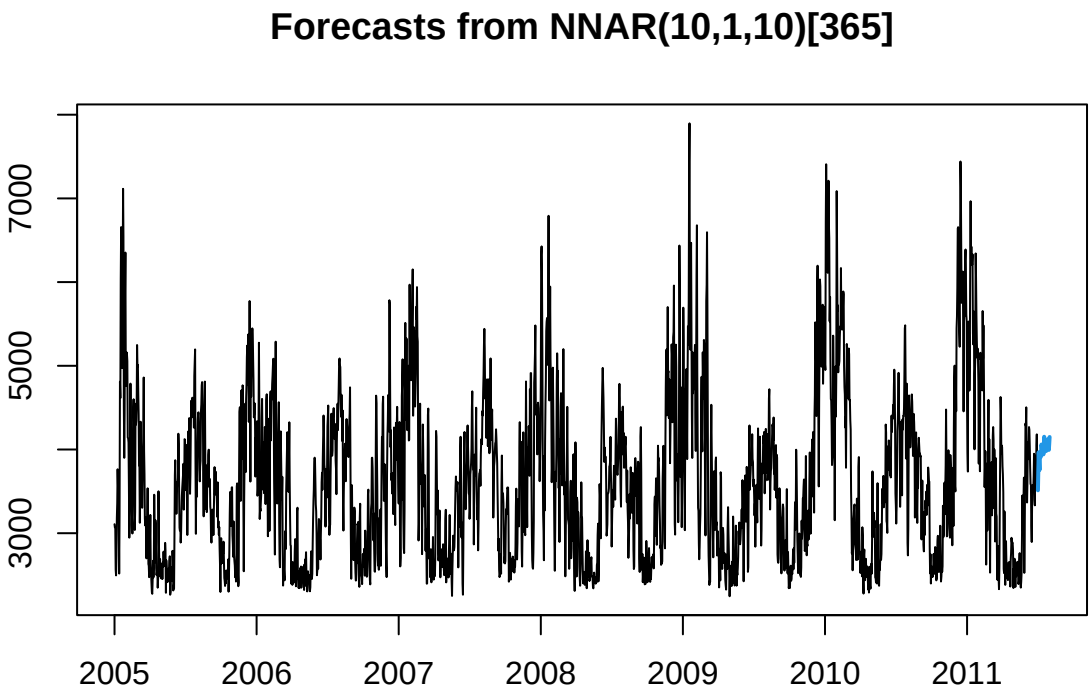

Figure 5: Neural Network, Version 2

```
#start with train window
temp<-temp_train
humidity<-humidity_train

ARIMAF_train1 <- auto.arima(ts_load_train,
                            seasonal=FALSE,
                            lambda=0,
                            xreg= cbind(fourier(ts_load_train, K=c(2,12)), temp, humidity))

#change to test window
temp <- temp_test
humidity <- humidity_test

ARIMAF_train_forecast1<- forecast(ARIMAF_train1,
                        xreg= cbind(fourier(ts_load_train, K = c(2,12), h = 31),
                               temp,
                               humidity), h=31)

accuracy(ARIMAF_train_forecast1$mean, ts_load_test)
```

```
##                ME     RMSE      MAE        MPE     MAPE      ACF1 Theil's U
## Test set 64.71734 685.8506 578.8199 -0.7295879 14.17149 0.6617674  1.488482
```

```
ARIMAF_1 <- auto.arima(msts_load,
                            seasonal=FALSE,
                            lambda=0,
                            xreg= cbind(fourier(msts_load, K=c(2,12)), temp, humidity))

ARIMAF_1_forecast<- forecast(ARIMAF_1,
```

```
                        xreg= cbind(fourier(msts_load, K = c(2,12), h = 31),
                               as.numeric(temp_July$mean),
                               as.numeric(humidity_July$mean)), h=31)

plot(ARIMAF_1_forecast)
```

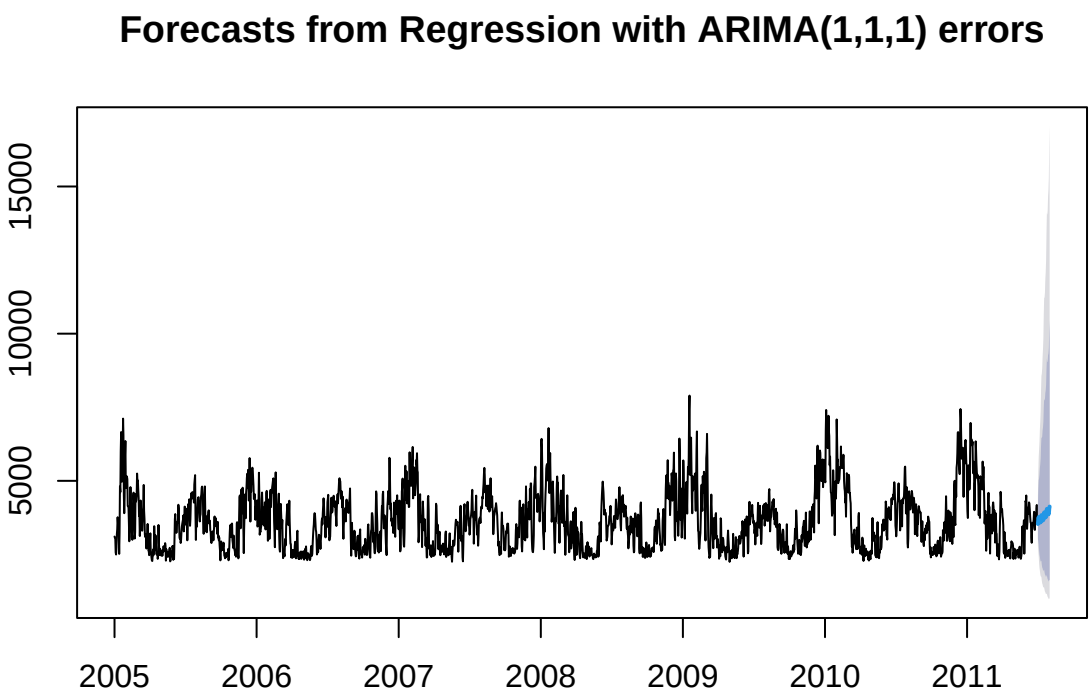

Figure 6: ARIMA+Fourier, Version 1

```
ARIMAF_1_data<- data.frame(
  date = seq(as.Date("2011-07-01"), length.out = 31, by = "day"),
  value = ARIMAF_1_forecast$mean
)

#write_csv(ARIMAF_1_data, 'Submissions/ARIMAF_1_data.csv')
```

A loop was run for k values between 1 and 3 for the weekly seasonality and between 8 and 15 for the annual seasonality. K = 2,10 yielded the best MAPE score and all AIC values were fairly similar. It did not however beat out the K = 2,12 after uploading to Kaggle. Therefore, the test window wasn't a perfect estimate of the actual forecasting window for model fitting.

```
## ARIMA +FOURIER (K = 2,10) Training Forecast & Test Window

#start with train window
temp<-temp_train
humidity<-humidity_train

ARIMAF_train2 <- auto.arima(ts_load_train,
                            seasonal=FALSE,
                            lambda=0,
                            xreg= cbind(fourier(ts_load_train, K=c(2,10)), temp, humidity))

#change to test window
```

```r
temp <- temp_test
humidity <- humidity_test

ARIMAF_train_forecast2<- forecast(ARIMAF_train2,
                        xreg= cbind(fourier(ts_load_train, K = c(2,10), h = 31),
                               temp,
                               humidity), h=31)

accuracy(ARIMAF_train_forecast2$mean, ts_load_test)
```

```
##                 ME     RMSE      MAE        MPE     MAPE      ACF1 Theil's U
## Test set 82.30431 676.8065 571.2475 -0.2630482 13.91268 0.6534017  1.449908
```

```r
ARIMAF_2 <- auto.arima(msts_load,
                             seasonal=FALSE,
                             lambda=0,
                             xreg= cbind(fourier(msts_load, K=c(2,10)), temp, humidity))

ARIMAF_2_forecast<- forecast(ARIMAF_2,
                        xreg= cbind(fourier(msts_load, K = c(2,10), h = 31),
                               as.numeric(temp_July$mean),
                               as.numeric(humidity_July$mean)), h=31)

plot(ARIMAF_2_forecast)
```

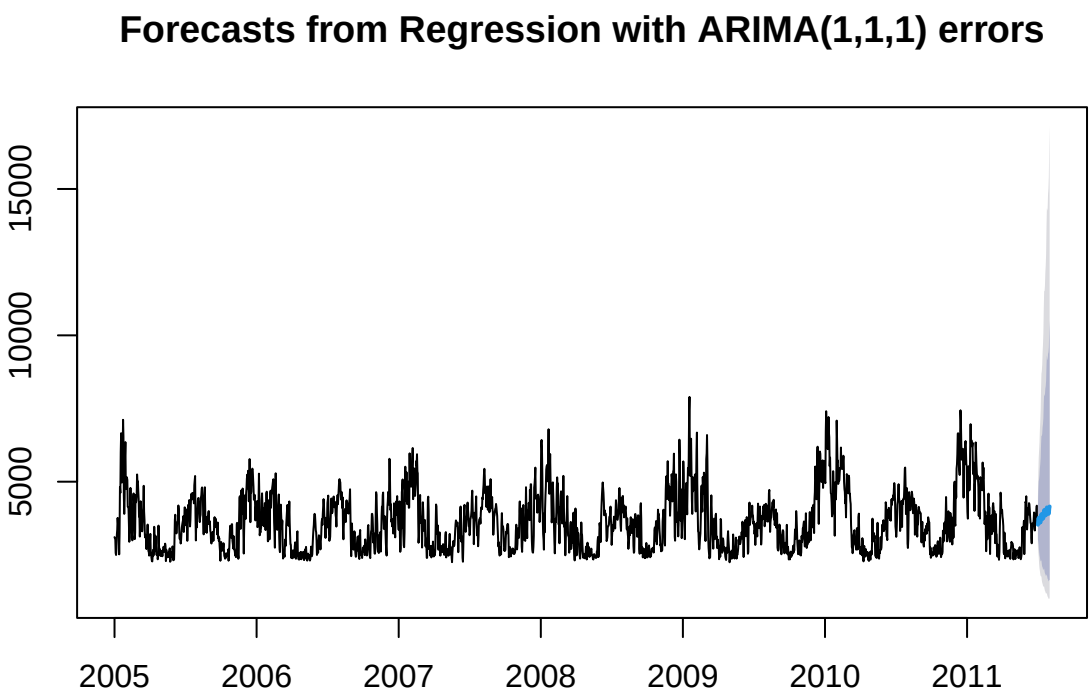

Figure 7: ARIMA+Fourier, Version 2

```r
ARIMAF_2_data<- data.frame(
  date = seq(as.Date("2011-07-01"), length.out = 31, by = "day"),
  value = ARIMAF_2_forecast$mean
)
```

```
#write_csv(ARIMAF_2_data, 'Submissions/ARIMAF_1_data.csv')
```

Summary of all models

Table 1: Forecast Accuracy for Electricy Load

	Name	July2011MAPE	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	TBATS	9.98	-92.44	554.79	444.17	-4.09	11.21	0.64	1.24
Test set1	NN K=2,12	11.34	34.50	327.54	251.65	0.06	6.04	0.31	0.68
Test set2	NN K=2,1	11.38	28.04	328.97	249.60	-0.20	6.03	0.38	0.68
Test set3	ARIMAF K=2,12	9.65	64.72	685.85	578.82	-0.73	14.17	0.66	1.49
Test set4	ARIMAF K=2,10	9.93	82.30	676.81	571.25	-0.26	13.91	0.65	1.45

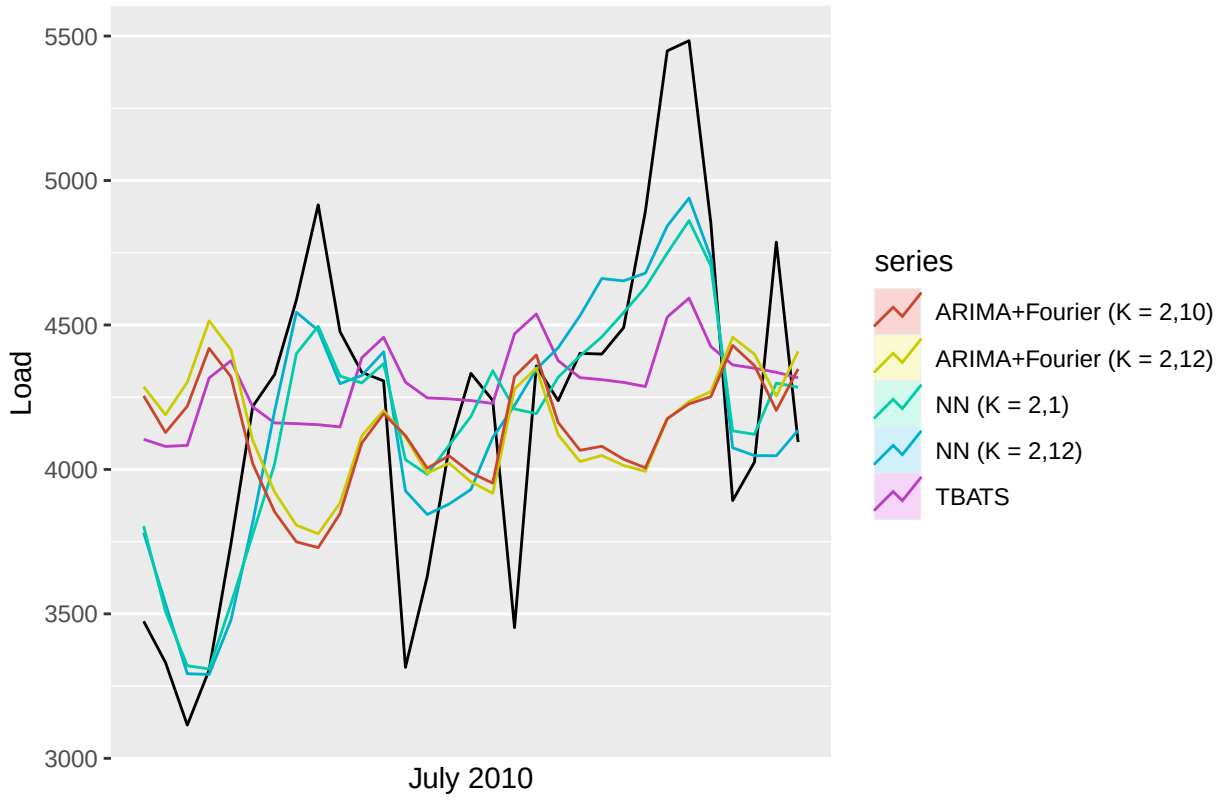

Figure 8: All Forecasts

Analysis and Conclusions

Using the training set up until June 30th, 2010 and comparing each models forecast for July 2010 to the actual data, the neural network model with terms K = 2,12 was the best fit with the lowest MAPE score. However, after using these models to forecast July 2011 using all data and uploading the forecast to Kaggle, the ARIMA + Fourier with terms K = 2,12 yielded the best MAPE score. This indicates that while training and testing models can help determine which model types work best, testing windows may not always estimate the best forecasting model to use. The data should be more deeply explored to determine if July 2010 was an anomaly year or if it displayed a level shift or other major change that would have significantly helped predict July 2011.